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Benjamin Mos
Southern Cross University

Marie Byrne
University of Sydney

Symon A. Dworjanyn
Southern Cross University

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Biogenic acidification reduces sea urchin gonad growth and increases susceptibility of aquaculture to ocean acidification

Benjamin Mos^{*1}, Maria Byrne² and Symon A. Dworjanyn¹

¹ National Marine Science Centre, Southern Cross University, PO Box 4321, Coffs Harbour, NSW 2450, Australia

² Schools of Medical and Biological Sciences, University of Sydney, Sydney, New South Wales, Australia

* Corresponding Author: B Mos

Email: benjamin.mos@scu.edu.au

HIGHLIGHTS:

- Sea urchin growth and gonads are reduced at high densities/ low seawater exchanges.
- CO₂ is increased and total alkalinity decreased in sea urchin culture water.
- These changes are caused by respiration and calcification = biogenic acidification.
- Reduced growth and gonad production was attributed to biogenic acidification.
- Low total alkalinity may leave aquaculture vulnerable to ocean acidification.

ABSTRACT:

Decreasing oceanic pH (ocean acidification) has emphasised the influence of carbonate chemistry on growth of calcifying marine organisms. However, calcifiers can also change carbonate chemistry of surrounding seawater through respiration and calcification, a potential limitation for aquaculture. This study examined how seawater exchange rate and stocking density of the sea urchin *Tripneustes gratilla* that were reproductively mature affected carbonate system parameters of their culture water, which in turn influenced growth, gonad production and gonad condition. Growth, relative spine length, gonad production and consumption rates were reduced by up to 67% by increased density (9 to 43 individuals.m⁻²) and reduced exchange rates (3.0 to 0.3 exchanges.hr⁻¹), but survival and food conversion efficiency were unaffected. Analysis of the influence of seawater parameters indicated that reduced pH and calcite saturation state (Ω_{Ca}) were the primary factors limiting gonad production and growth. Uptake of bicarbonate and release of respiratory CO₂ by *T. gratilla* changed the carbonate chemistry of surrounding water. Importantly total alkalinity (A_T) was reduced, likely due to calcification by the urchins. Low A_T limits the capacity of culture water to buffer against acidification. Direct management to counter biogenic acidification will be required to maintain productivity and reproductive output of marine calcifiers, especially as the ocean carbonate system is altered by climate driven ocean acidification.

KEYWORDS: calcite saturation, calcification, density, seawater exchange rate, nitrogenous wastes, sea urchin, *Tripneustes gratilla*

INTRODUCTION

The biological effects of anthropogenically-driven decreases in oceanic pH (ocean acidification) highlights how changes in seawater carbonate chemistry (pH, partial pressure of dissolved carbon dioxide [$p\text{CO}_2$] and calcite saturation state [ΩCa]) can reduce growth and fitness of calcifying organisms (Kroeker et al., 2013; Kroeker et al., 2010). Marine calcifiers can also alter carbonate chemistry through respiration, photosynthesis, calcification and dissolution of calcium carbonate (Anthony et al., 2011; Hurd et al., 2011; Schneider et al., 2011). In natural systems with low water exchange these processes can create greater changes in carbonate chemistry than that predicted for near future ocean acidification, as shown for boundary layers in algal beds and intertidal rock pool habitats (Hurd et al., 2011; Truchot and Duhamel-Jouve, 1980).

In closed aquaculture systems, acidification and reduced buffering capacity (low total alkalinity [A_T]) of culture water is also a serious limitation for aquaculture of calcifying organisms (Barton et al., 2012) as economics demand the highest possible densities of animals with the lowest practical water exchange rates (Mos et al., 2015). However, the extent of the effect of intensive production of calcifying organisms on the carbonate chemistry of their culture water and the potential for this to limit production of these organisms is poorly understood (Mos et al., 2015). It is critical to address this knowledge gap to predict how elevated atmospheric CO_2 levels and associated impacts on carbonate chemistry of the oceans may impact aquaculture production and to increase the efficiency and sustainability of the culture of marine calcifiers.

The extent to which calcifying organisms alter carbonate chemistry in aquaculture systems depends on the intensity of production and the rate of water turnover (Mos et al., 2015).

Density and growth rates of organisms determine respiratory CO_2 production and the

depletion of carbonate by calcification, whilst removal of CO₂ and replenishment of carbonate is governed by water turnover. Capacity to perturb carbonate systems is also likely to vary across life-history stages. For example, rapid calcification by fast-growing juveniles reduces A_T, limiting the capacity of their culture water to buffer changes in carbonate chemistry due to excretion of respiratory CO₂ (Mos et al., 2015), which is also high in juveniles (Mos et al., 2015). In contrast, adult calcifiers may have comparatively smaller effects on the A_T of culture water because their growth rates (and therefore calcification rates) are inherently lower (Schnute, 1981) or because they may allocate more effort to reproductive rather than somatic growth (Otero-Villanueva et al., 2004; Russell, 1998).

The impact of variation in seawater carbonate chemistry on organisms also changes across life history stages. Fertilisation and early embryos of marine invertebrates are generally robust to seawater acidification, but later calcifying larval stages are generally highly sensitive (Byrne and Przeslawski, 2013). The growth of newly metamorphosed (Wolfe et al., 2013) and juvenile (Albright et al., 2012; Asnaghi et al., 2014; Courtney et al., 2013; Holtmann et al., 2013; Mos et al., 2015) sea urchins is reduced under low pH/ high pCO₂ conditions. However, little is known about how seawater acidification may affect the development of reproductive structures (Dupont et al., 2013; Kurihara et al., 2013; Siikavuopio et al., 2007b; Stumpp et al., 2012b; Suckling et al., 2015; Uthicke et al., 2014; Uthicke et al., 2013). This is a particularly important gap in our knowledge of the effects of acidification on the aquaculture of sea urchins because they are grown for their gonads.

This study investigated the impact of changes in seawater carbonate chemistry in culture systems during the period of growth to reproductive maturity of *Tripneustes gratilla*, a fast-growing, high value tropical sea urchin targeted for commercial aquaculture (Dworjanyn and Pirozzi, 2008; Dworjanyn et al., 2007; Mos et al., 2015; Mos et al., 2011; Scholtz et al., 2013;

Seymour et al., 2013) and biocontrol (Westbrook et al., 2015). Specifically, we examined the effects of stocking density and seawater exchange rate on seawater physico-chemical parameters and, in turn, growth, survival, relative spine length, gonad production and gonad condition of the sea urchins. Data from three density and three exchange rate treatments were assessed with respect to temperature, pH, salinity, A_T , nitrate, DO, pCO_2 and calcite saturation state (Ω_{Ca}) in the culture system. We predicted that biogenic alteration of carbonate system parameters would have a major influence on growth and gonad production in *T. gratilla*, with implications for the culture of this and other calcifying organisms in the face of an acidifying ocean.

MATERIALS AND METHODS

Study Organism

Tripneustes gratilla were cultured at the National Marine Science Centre, Southern Cross University, Coffs Harbour, Australia (30°12.5'S, 153°16.1'E) using established protocols (Mos et al., 2015; Mos et al., 2011) and gametes from five males and five females. Larvae were induced to settle using naturally derived biofilms and *Sargassum* sp. conditioned seawater in a flow-through seawater system at 24 – 25 °C and 35 ppt salinity. Settled juveniles consumed the biofilm until a test diameter (TD) of 5 – 10 mm when they were fed a seaweed diet (*Sargassum* sp.). *T. gratilla* were used in the experiment at the size when they began to produce gonads (~ 70 mm TD, approx. 7 months of age) (Mos et al., 2015). They were kept at 26 – 27 °C and not fed for two weeks prior to the experiment to standardise their nutritional condition.

Culture System

A fully factorial design was used to test the effects of three densities, low, medium and high (1, 3 and 5 individuals.replicate⁻¹ respectively) and three seawater exchange rates, low,

medium and high (0.3, 1.0 and 3.0 exchanges.hr⁻¹ respectively) on the growth, gonad production, gonad condition, consumption rates, FCE and survival of *T. gratilla* over six weeks. Initial densities were equivalent to 9, 26 and 43 individuals.m⁻² (also see Fig. S1). Each treatment had five replicates consisting of 5 L rearing containers (180 mm Ø base, 200 mm H) with over-flow holes that maintained the water volume at 4 L. Replicates were randomly assigned positions in a water bath (100 mm deep) to maintain stable water temperatures. Filtered (20 µm) seawater was supplied from an 800 L header tank, heated (26 – 27 °C, optimal for growth and reproductive maturation (Mos et al., 2015)) using a looped flow-through 3000 Watt heater (Elecro Engineering TO3 Ti). Seawater exchange rates for each replicate were regulated using irrigation drip taps. Replicates were individually aerated (0.71 L.min⁻¹, regulated using flow control valves) and maintained under a 12:12 photoperiod ('cool white' fluorescent). Sea urchins were fed daily *ad libitum* with the seaweed *Sargassum* sp. which is a high preference food (Dworjanyn et al., 2007; Seymour et al., 2013).

Effects of Density and Seawater Exchange Rate on Seawater Parameters

Daily temperature, pH_{NIST} and salinity measurements were recorded for each replicate and the ambient seawater supplied to the system using Hach[®] HQ40d multi-controller, Hach[®] PHC101 pH probe and Hach[®] CDC101 conductivity probe. Dissolved oxygen (DO) was measured three times a week (Hach[®] LDO101). Nitrite, nitrate and TAN (Total Ammonia Nitrogen) were measured from water samples collected daily, pooled from equal volumes taken from replicates in each treatment and from the ambient seawater supplied to the system (Palintest[®] photometer 7100, Ammonia AP152, Nitricol AP109 and Nitrate AP163). Total alkalinity (A_T) was measured in water samples (100 mL) collected weekly, pooled from equal volumes taken from replicates in each treatment and the ambient seawater supplied to the experiment respectively. Samples were filtered (0.45 µm) and fixed with 5 µL of saturated

HgCl₂. A_T was determined by potentiometric titration using a Metrohm 888 Titrando and certified reference standards (Dickson et al., 2007). Values for pCO₂ and ΩCa were calculated from weekly mean temperature, pH, salinity and A_T data using CO2SYS (Pierrot et al., 2006) using the dissociation constants of Mehrbach (1973) as refitted by Dickson and Millero (1987).

Estimates for the amount of CO₂ produced by *T. gratilla* during week six were compared to pCO₂ levels in the culture water (Table S4). The amount of CO₂ produced was estimated using regression coefficient (b) and constant (K) values for *T. gratilla* (Dy et al., 2002) under similar temperature and salinity conditions to this study, for the equation $M = KW^b$ relating body mass and metabolic rate (Webb et al., 1977).

Effects of Density and Seawater Exchange Rate on Growth and Survival

None of the urchins in the experiment died. To assess growth, test diameter (TD) and wet weight (WW) measurements were taken weekly using standard techniques (Dworjanyn et al., 2007; Mos et al., 2015; Watts et al., 2010). Urchins were weighed to the nearest 0.01 g after removal of excess seawater by placing them on dry paper towel for 30 sec. Image analysis software, ImageJ (NIH, USA) was used to measure TD from digital photographs of the aboral side of the urchins. Two perpendicular measurements at the longest axis were taken for each urchin and the average used for statistical analysis. There was no significant difference in the initial TD ($F_{8, 44} = 1.54$, $P = 0.1769$; Mean = 71.1 mm ± 0.4 SE) or WW ($F_{8, 44} = 0.87$, $P = 0.5547$; Mean = 107.7 g ± 1.6 SE) of *T. gratilla* among treatments. After six weeks, Specific Growth Rate (SGR) and Linear Growth Rate (LGR) were calculated using standard formulae:

$$\text{SGR (\%)} = ([\ln W_f - \ln W_i] \times 100) / t_f - t_i$$

$$\text{LGR} = (L_f - L_i) / t_f - t_i$$

where W_i and W_f were the initial and final mean WW (g) respectively of the urchins, L_i and L_f were the initial and final mean TD (mm) respectively of the urchins, and t_i and t_f were the initial and final time points (weeks).

Spine length to TD ratios (relative spine length) were calculated using Feret's Diameters obtained from digital photographs of the aboral sides of the urchins (Mos et al., 2015):

$$\text{Relative spine length} = \frac{1}{2}(FD_S - FD_T) / FD_T \times 100$$

where FD_S and FD_T were Feret's Diameters of circles encompassing the urchins' spines and tests respectively. There was no difference in the initial relative spine length of *T. gratilla* among treatments ($F_{8, 44} = 1.05$, $P = 0.4177$; Mean = 16.7% of TD $\pm$ 0.5 SE).

Effects of Density and Seawater Exchange Rate on Consumption and FCE

Consumption rates were assessed on a daily basis for seven days (week 4 – 5) by recording initial and final WW of *Sargassum* sp. fed to each replicate and the elapsed time. Autogenic change in WW of *Sargassum* sp. was assumed to be negligible because a pilot trial without sea urchins (total $n = 27$, 9×3 seawater exchange rates) showed autogenic changes were $< \pm 0.8\%$ and urchins always consumed $> 50\%$ of *Sargassum* sp. available daily. Consumption rates and FCEs were calculated using standard formulae (Mos et al., 2015):

$$\text{Consumption} = [(S_i - S_f) / d_i] / (t_f - t_i)$$

where S_i and S_f were the initial and final WW respectively of the *Sargassum* sp., d_i was the initial mean WW of the urchins and t_f and t_i were the final and initial time points respectively.

$$\text{Food Conversion Efficiency (\%)} = [(d_f - d_i) / S] \times 100$$

where d_i and d_f were the initial and final mean WW respectively of the urchins and S was the total WW of *Sargassum* sp. consumed.

Effects of Density and Seawater Exchange Rate on Gonad Production

Initial gonad index (GI) was assessed from 25 randomly selected urchins using standard techniques and formula (Seymour et al., 2013).

$$GI = (W_G / W_U) \times 100$$

where W_G was the WW (g) of the gonads and W_U was the WW (g) of the urchin. After six weeks, all urchins in the experiment were dissected to obtain GIs. Gonads were photographed (Panasonic Lumix DMC-FT10) within a light box alongside a paint colour series: white (Polar cape, Wattyl A10w), tan (Apricotta, Wattyl B75w), pale orange (Delicious Peach+, Wattyl B76w), mid orange (Mandarin Grove+, Wattyl C22w), intense orange (Tangella+, Wattyl B79w) (Fig. 3a). White balance was set to *auto* and digital images were stored as JPEG files (resolution 4320×3240 pixels, three channels of 24 bit RGB colour information). Gonad colour was subjectively quantified by one author (BM) using a rating scale (1 – 5) corresponding to the nearest colour in the paint colour series (1 representing white, 5 intense orange, Fig. 3a). Gonad colour was also objectively quantified using Strength of red (S_{Red}). S_{Red} were calculated from RGB digital numbers following Gillespie et al. (1987) measured from selected portions of the images (i.e. gonads only) using ImageJ (NIH). High S_{Red} ratios corresponded with greater orange colouration, which is indicative of high gonad quality and market value (Agatsuma et al., 2005; Pearce et al., 2002).

One gonad from each urchin was fixed in Bouin's fixative (Sigma-Aldrich HT10132) and then stored in 70% ethanol. Five representative gonads from individuals in the high density/low exchange rate and the low density/high exchange rate treatments respectively were processed for routine wax histology. The gonad sections (7 μ m thick) were stained in haematoxylin and eosin. Under microscopic examination (40 – 100 $\times$ magnification), condition of the gonads was determined by the extent and density of eosinophilia which

indicates the amount of nutrients stored in the nutritive phagocytes, the cells that dominate the gonad lumen of well-fed urchins prior to gamete maturation (Byrne, 1990).

Statistical Analysis

Data for growth, survival, relative spine length, consumption, FCE and seawater physico-chemical parameters (pH, $p\text{CO}_2$, ΩCa , A_T) were analysed using permutational analysis of variance (PERMANOVA, Anderson, 2001) with seawater exchange rate and density as orthogonal fixed factors in a fully-crossed design. Containers were used as the level of replication.

Multiple linear regressions were used to examine the relationships between water quality parameters (temperature, DO, salinity, pH, $p\text{CO}_2$, ΩCa) and SGR, LGR, relative spine length, consumption rates, FCE, GI and gonad condition (colour rating and S_{Red}). Mean growth (SGR, LGR, relative spine length), gonad production (GI) and gonad condition (colour rating, S_{Red}) in each replicate over the six week experiment ($n = 5$ per treatment) were compared to mean seawater parameters in each replicate over the six week experiment as gonad characteristics were only measured at the end of the experiment and seawater parameters remained relatively stable throughout the experiment (Fig. 4, Table S3). To include A_T in models, temperature, DO, salinity, natural log transformed pH, $p\text{CO}_2$, ΩCa , LGR, SGR and relative spine length were averaged across weeks within each treatment to match the sampling design for A_T ($n = 6$ per treatment). However A_T contributed little to the models and the models produced were similar to those for parameters averaged for each replicate over time (Table 2). Mean consumption rates for each replicate were compared to DO, temperature, salinity, natural log transformed pH, $p\text{CO}_2$ and ΩCa values in each replicate during week 4 – 5 of the experiment. Marginal tests were used to test the relationship between a single seawater parameter and growth, gonad production or gonad condition, and

check for collinearity amongst predictors. Seawater parameters with P values > 0.01 were excluded from further analysis.

Linear regressions were conducted using the distance-based linear modelling (DISTLM) with distance based redundancy analysis (dbRDA) routine of Primer 6 (Primer-E, Plymouth) with PERMANOVA⁺ extension (v.6.1.11) software (Anderson, 2001; Anderson et al., 2008; McArdle and Anderson, 2001). The BEST model selection routine with AICc selection criterion (a modified version of Akaike's Information Criterion), based on 9999 permutations, was used to select the five best fitting models.

RESULTS

Effects of Density and Seawater Exchange Rate on Growth and Survival

Linear (LGR) and Specific (SGR) Growth Rates of *T. gratilla* ranged from 1.30 – 3.84 mm.week⁻¹ and 6.03 – 13.90 %.week⁻¹, respectively (Fig. 1a,b). Both LGR and SGR were dependent on a significant interaction between seawater exchange rate and density (Fig. 1a,b Table 1). In general, there were higher growth rates in high exchange rate/low density treatments and lower growth rates in low exchange rate/high density treatments (Fig. 1a, see Table 1 for a full description of the interaction).

Relative spine lengths of *T. gratilla* ranged from 9 – 18% of TD (Fig. 1c) and depended on seawater exchange rate and density with no interaction between treatments (Fig. 1c, Table 1). Relative spine lengths were reduced by decreased seawater exchange rate (low < medium < high) and increased density (low > medium > high).

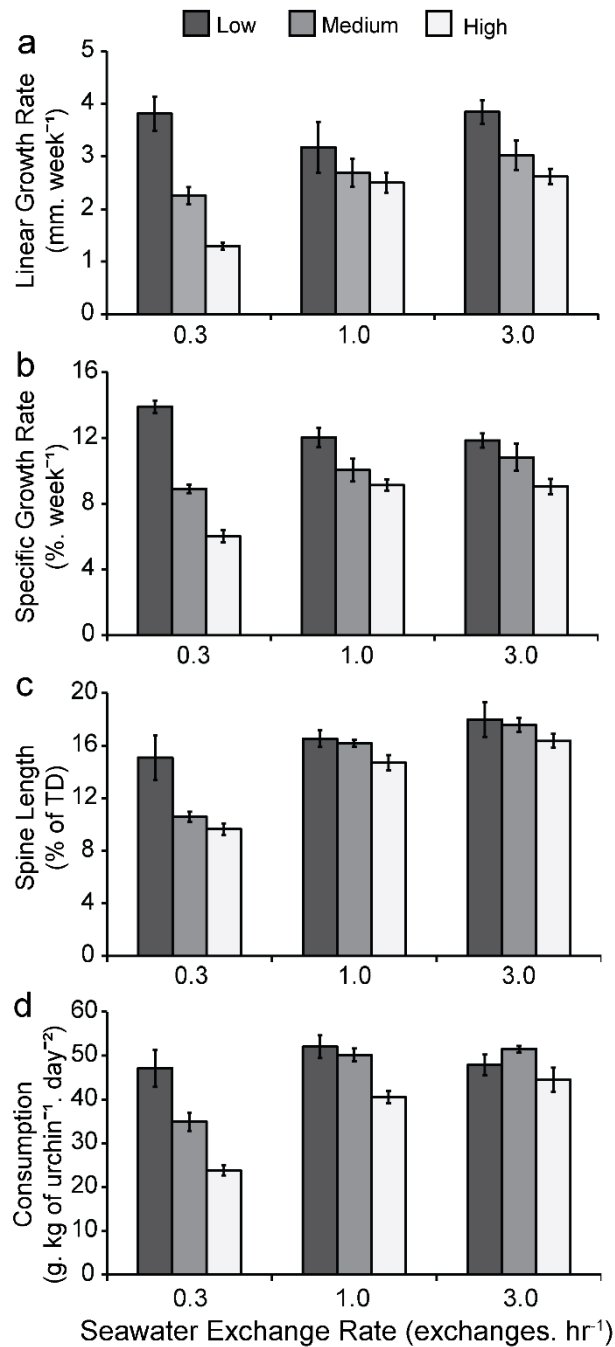


Figure 1: Effects of density and seawater exchange rate on the growth of *Tripneustes gratilla* after six weeks and consumption rates over seven days in a flow-through seawater system. (a) Linear Growth Rate (LGR), (b) Specific Growth Rate (SGR), (c) relative spine length (ratio of spine length to test diameter), (d) consumption rates. Stocking density: dark bars = Low (9 individuals.m⁻²); medium bars = Medium (26 individuals.m⁻²); light bars = High (43 individuals.m⁻²). (a,b,d) LGR, SGR and consumption rates were dependent on a significant interaction between seawater exchange rate and density (Table 1). (c) Relative spine length was dependent on seawater exchange rate (Table 1, 0.3 < 1.0 < 3.0) and density (Table 1, Low > Medium > High). Data are means $\pm$ S.E., n = 5.

Table 1: PERMANOVA analyses examining the effects of density and seawater exchange rate on linear growth rate (LGR), specific growth rate (SGR), relative spine length (ratio of spine length to test diameter), gonad index (GI), subjective gonad colour (GC) and Strength of red in RGB digital images of gonads (S_{Red}) for *Tripneustes gratilla* after six weeks in a flow-through seawater system and consumption rate and food conversion efficiency (FCE) of *T. gratilla* over 7 days in a flow-through seawater system.

	Source	df	MS	F	P	Post-hoc tests	
LGR	Exchange Rate	2	1.88	5.38	0.0090	<i>Interaction</i>	
	Density	2	8.39	23.97	0.0001	<i>Interaction</i>	
	Er × De	4	1.13	3.22	0.0225	low (3.0 = 1.0 = 0.3)	0.3 (low > medium > high)
	Residual	36	0.35			medium (3.0 > 0.3)	1.0 (low = medium = high)
						high (3.0 = 1.0 > 0.3)	3.0 (low > high)
SGR	Exchange Rate	2	3.95	3.05	0.0569	<i>Interaction</i>	
	Density	2	77.65	59.94	0.0001	<i>Interaction</i>	
	Er × De	4	11.39	8.80	0.0002	low (3.0 = 1.0 > 0.3)	0.3 (low > high)
	Residual	36	1.30			medium (3.0 = 1.0 = 0.3)	1.0 (low > medium > high)
						high (3.0 = 1.0 > 0.3)	3.0 (low > high)
Spine Length	Exchange Rate	2	122.65	35.07	0.0001	3.0 > 1.0 > 0.3	
	Density	2	33.26	9.51	0.0008	low > medium > high	
	Er × De	4	8.55	2.45	0.0634		
	Residual	36	3.50				
GI	Exchange Rate	2	16.09	13.47	0.0001	3.0 = 1.0 > 0.3	
	Density	2	15.57	13.04	0.0002	low = medium > high	
	Er × De	4	0.40	0.34	0.8517		
	Residual	36	1.19				
GC	Exchange Rate	2	0.97	6.53	0.0035	3.0 = 1.0 > 0.3	
	Density	2	4.09	27.46	0.0001	low > medium > high	
	Er × De	4	8.41E-6	0.56	0.6906		
	Residual	36	0.15				
S_{Red}	Exchange Rate	2	2.14E-3	3.40	0.0424	3.0 > 0.3	
	Density	2	3.10E-3	4.92	0.0104	low > medium = high	
	Er × De	4	1.40E-3	2.22	0.0857		
	Residual	36	6.29E-4				
Consumption	Exchange Rate	2	7.76E-4	28.87	0.0001	<i>Interaction</i>	
	Density	2	6.48E-4	24.12	0.0001	<i>Interaction</i>	
	Er × De	4	1.40E-4	5.22	0.0031	low (3.0 = 1.0 = 0.3)	0.3 (low > medium > high)
	Residual	36	2.69E-5			medium (3.0 = 1.0 > 0.3)	1.0 (low = medium > high)
						high (3.0 = 1.0 > 0.3)	3.0 (medium > high)
FCE	Exchange Rate	2	92.74	2.90	0.0628		
	Density	2	33.87	1.06	0.3634		
	Er × De	4	4.11	0.13	0.9763		
	Residual	36	31.93				

Er = Exchange Rate; De = Density; Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square. Post-hoc tests: low, medium and high correspond to low, medium and high density treatments respectively. 0.3, 1.0 and 3.0 correspond to seawater exchange rate treatments (0.3, 1.0 and 3.0 exchanges.hr⁻¹ respectively).

Effects of Density and Seawater Exchange Rate on Consumption and FCE

Urchins consumed 2 – 5% of their body mass per day (Fig. 1d). Consumption rates were dependent on a complex interaction between seawater exchange rate and density, but in

general consumption was higher in high exchange rate/low density treatments and lower in low exchange rate/high density treatments (Fig. 1d, Table 1). Food conversion efficiency (FCE) ranged from 22 – 31%, but was not significantly different between density or seawater exchange rate treatments (Table 1).

Effects of Density and Seawater Exchange Rate on Gonad Production

The gonad indices (GI) of *T. gratilla* after six weeks were higher than in initial controls ($1.43\% \pm 0.21$ S.E.) in all treatments and depended on seawater exchange rate and density with no interaction between treatments (Fig. 2a, Table 1). GIs were reduced by increased density (low > medium = high) and were lowest in the low seawater exchange rate (low < medium = high).

Subjective and measured (S_{Red}) gonad colour was dependent on seawater exchange rate and density with no interaction between treatments (Fig. 2b,c, Table 1). These ratings were reduced by increased density (Subjective: low > medium > high; S_{Red} : low > medium = high) and lower seawater exchange rate (Subjective: low < medium = high; S_{Red} : low = medium, medium = high, low < high).

Gonads from the high density/low exchange rate treatment were substantially smaller and paler in colour than gonads from the low density/high exchange rate treatment (Fig. 3a).

Histological condition of the gonads also differed among treatments (Fig. 3b,c). Gonads from the high density/low exchange rate treatment were in poor condition and had few nutritive cells and low nutrient storage levels, as indicated by an empty lumen and the pale or no staining with eosin (Fig. 3b). In contrast, gonads from the low density/high exchange rate treatment were dense, with greater numbers of nutritive cells and stored nutrients that conveyed strong eosinophilia to the sections (Fig. 3c).

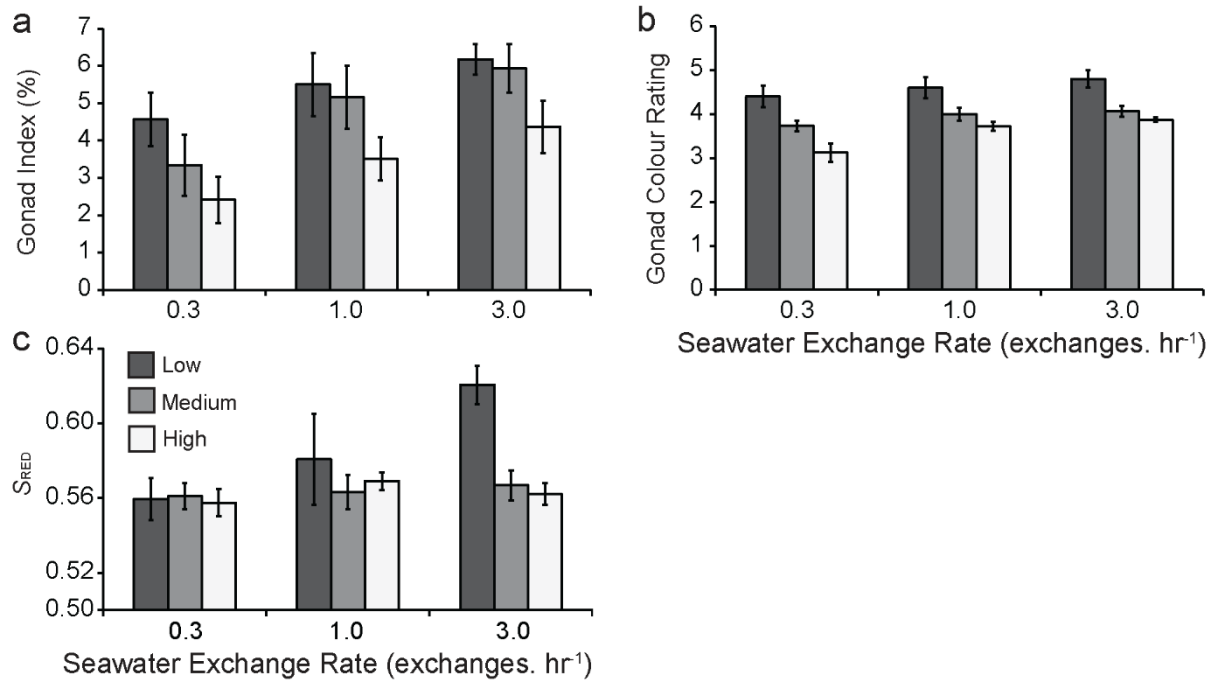


Figure 2: Effects of density and seawater exchange rate on gonad production and gonad colour of *Tripneustes gratilla* after six weeks in a flow-through seawater system. (a) Gonad Index. (b) Subjective Gonad Colour Rating (1 – 5). (c) Strength of Red (S_{RED}) measured from digital images of the gonads. Stocking density: dark bars = Low (9 individuals.m⁻²); medium bars = Medium (26 individuals.m⁻²); light bars = High (43 individuals.m⁻²). (a) Gonad index (GI) was dependent on seawater exchange rate (Table 1, 0.3 < 1.0 = 3.0) and density (Table 1, low > medium > high). (b) Subjective gonad colour rating was dependent on seawater exchange rate (Table 1, 0.3 < 1.0 = 3.0) and density (Table 1, Low > Medium > High). (c) S_{RED} was dependent on seawater exchange rate (Table 1, 0.3 = 1.0, 1.0 = 3.0, 0.3 < 3.0) and density (Table 1, Low > Medium = High). Data are means $\pm$ S.E., n = 5.

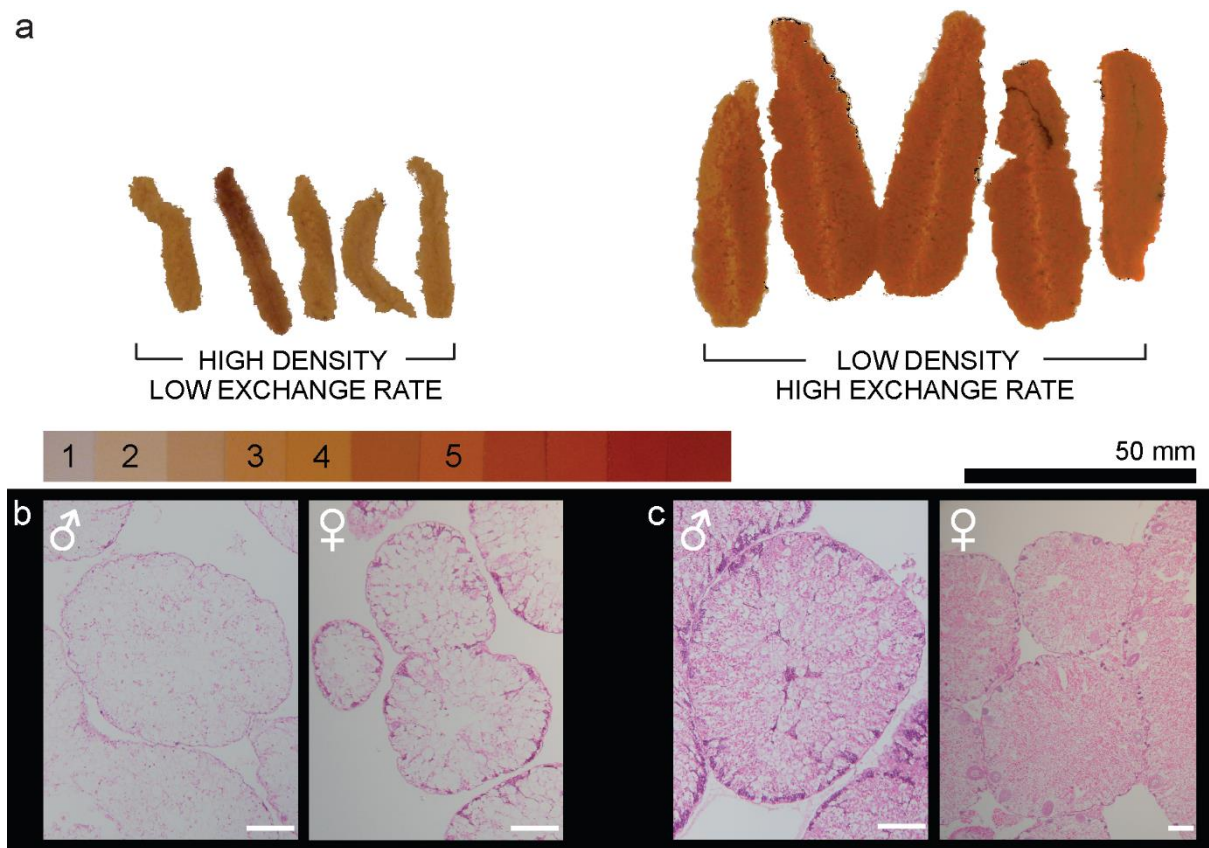


Figure 3: The effects of density and seawater exchange rate on the condition of the gonads of *Tripneustes gratilla* after six weeks in a flow-through seawater system. (a) Representative gonad size and colour of *T. gratilla* in high density/low exchange rate and low density/high exchange rate treatments respectively after six weeks. (b - c) Gonad histology of male (♂) and female (♀) *T. gratilla* from the high density/low exchange rate treatment (b) and the low density/high exchange rate treatment (c) after six weeks showing the higher density of nutritive cells (eosin positive) in the latter. (a) Numbers on the colour series represent the five colours used to assess subjective gonad colour rating. (b - c) Scale bars are 100 µm.

Effects of Density and Seawater Exchange Rate on Seawater Parameters

Seawater pH was lower in all treatments (pH_{NIST} 7.63 – 8.08) compared to the ambient seawater (8.10) supplied to the culture system (Fig 4a). pH was dependent on a significant interaction between seawater exchange rate and density (Fig. 4a, Table S2). In general, pH was lower in high density/low exchange rate treatments than in low density/high exchange rate treatments. The effect of density on pH was greater in the low exchange rate treatment

than in the medium exchange rate treatment which was greater than in the high exchange rate treatment (Fig. 4a).

Carbon dioxide levels ($p\text{CO}_2$) were higher in experimental treatments (342 – 715 μatm) compared to ambient seawater supplied to the culture system (322 μatm) (Fig. 4b). $p\text{CO}_2$ was dependent on a complex interaction between seawater exchange rate and density (Fig. 4b, Table S2). In general, $p\text{CO}_2$ was higher in high density/low exchange rate treatments than in low density/high exchange rate treatments.

Mineral saturation (ΩCa) was lower in all treatments (1.52 – 5.94) compared to the ambient seawater supplied to the culture system (6.31) and was dependent on a significant interaction between exchange rate and density (Fig. 4c, Table S2). In general, ΩCa was lower in high density/low exchange rate treatments than in low density/high exchange rate treatments. The effect of density on ΩCa was generally greater in the low exchange rate treatment than in the medium exchange rate treatment which was greater than in the high exchange rate treatment (Fig. 4c).

Total alkalinity (A_T) was lower in all treatments (1363 – 2281 $\mu\text{mol.L}^{-1}$) compared to the ambient seawater supplied to the culture system (2303 $\mu\text{mol.L}^{-1}$) and was dependent on a complex interaction between exchange rate and density (Fig. 4d, Table S2). In general, A_T was lower in high density/low exchange rate treatments than in low density/high exchange rate treatments.

Mean values for temperature, DO, salinity and nitrate concentration recorded in each of the treatments are reported in Table S3. TAN and nitrite were not detected ($< 0.00 \text{ mg.L}^{-1}$).

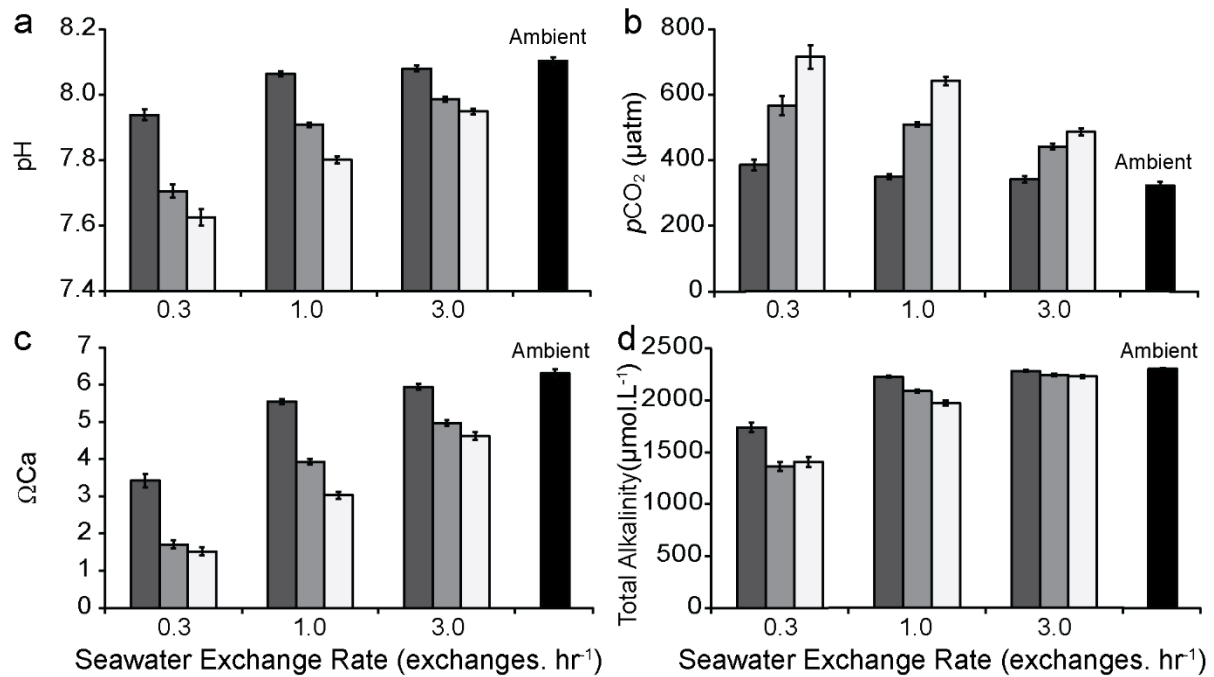


Figure 4: Effects of seawater exchange rate and density of *Tripneustes gratilla* on water quality parameters in a flow-through seawater system over six weeks. (a) pH_{NIST}, (b) partial pressure of dissolved carbon dioxide ($p\text{CO}_2$), (c) calcite saturation state (ΩCa), (d) total alkalinity (A_T). Stocking density: dark bars = Low (9 individuals.m⁻²); medium bars = Medium (26 individuals.m⁻²); light bars = High (43 individuals.m⁻²). Ambient denotes values in the seawater supplied to the system (data are means $\pm$ S.E. of six weekly averages). (a,b,c,d) pH, $p\text{CO}_2$, ΩCa and A_T were dependent on a significant interaction between seawater exchange rate and density (Table S2). Data are means $\pm$ S.E., $n = 5$ (pH, $p\text{CO}_2$, ΩCa) or 6 (A_T).

Effects of Environmental Parameters on Growth, Consumption Rate and Gonad

Production

No single model of the relationship between seawater parameters and SGR, LGR, relative spine length, GI or consumption rates was supported by the data as all AICc values in the top five models were similar (Table 2). However all the top five models contained the same carbonate chemistry parameters (Table 2, Table S1). Other parameters (salinity, temperature, DO) appeared to contribute little to the models. SGR, LGR and GI were correlated with ΩCa and pH (Table 2). Relative spine length was correlated with pH and $p\text{CO}_2$ (Table 2).

Consumption rates were correlated with ΩCa , pH and $p\text{CO}_2$ (Table 2).

Table 2: Model selection to estimate factors influencing the Specific Growth Rate (SGR), Linear Growth Rate (LGR), relative spine length (RSL), gonad index (GI) and consumption rates of *Tripneustes gratilla* cultured under all combinations of three densities and three seawater exchange rates. In each case, ‘predictors’ refers to the combination of explanatory variables included in the model. $\Delta AICc$ is the difference in AICc between the model and the best fitting model. AIC-weight ($AICc_{wt}$) is the probability that this model represents the best-fitting model among those considered. ‘Sum of $AICc_{wt}$ ’ refers to the sum of $AICc_{wt}$ scores for the given model and all better-fitting models. For each dataset, the five best models are presented. ΩCa = Calcite Saturation State, DO = dissolved oxygen, pCO_2 = dissolved carbon dioxide, Sal. = Salinity, Temp. = Temperature.

Dataset	Predictors	AICc	$\Delta AICc$	$AICc_{wt}$	Sum of $AICc_{wt}$	R^2
SGR	ΩCa , pH, DO	31.74	0	0.47	0.47	0.71
	ΩCa , pH	32.9	1.16	0.27	0.74	0.68
	ΩCa , pH, DO, pCO_2	33.89	2.15	0.16	0.9	0.71
	ΩCa , pH, pCO_2	35.24	3.5	0.08	0.98	0.68
	pCO_2 , DO	38.3	6.56	0.02	1	0.64
LGR	ΩCa , pH	-25.2	0	0.45	0.45	0.47
	ΩCa , pH, DO	-23.01	2.19	0.15	0.6	0.47
	pCO_2	-23.01	2.19	0.15	0.75	0.41
	ΩCa , pH, pCO_2	-22.9	2.3	0.14	0.89	0.47
	pH, DO	-22.3	2.9	0.11	1	0.44
RSL	pH, Sal.	55.16	0	0.25	0.25	0.72
	pH, pCO_2	55.18	0.02	0.25	0.5	0.72
	pH, pCO_2 , ΩCa , Temp.	55.72	0.56	0.19	0.68	0.75
	pH, pCO_2 , ΩCa	55.82	0.66	0.18	0.85	0.73
	pH, Sal., ΩCa , Temp.	56.29	1.13	0.14	1	0.74
GI	pH	10.8	0	0.36	0.36	0.52
	ΩCa	11.96	1.16	0.2	0.56	0.51
	ΩCa , DO	12.1	1.3	0.19	0.74	0.53
	ΩCa , pCO_2	12.77	1.97	0.13	0.88	0.52
	ΩCa , pH	12.92	2.12	0.12	1	0.52
Consumption	ΩCa , pH, pCO_2	166.98	0	0.49	0.49	0.66
	ΩCa , pH, pCO_2 , Sal.	169.08	2.1	0.17	0.66	0.66
	ΩCa , pH, pCO_2 , DO	169.38	2.4	0.15	0.81	0.66
	ΩCa , pH, pCO_2 , Temp.	169.52	2.54	0.14	0.95	0.66
	ΩCa , pH, Sal.	171.52	4.54	0.05	1	0.62

DISCUSSION

Biogenic acidification of culture water exacerbated by high stocking densities and low seawater exchange, reduced gonad development and quality, and growth of *T. gratilla*. This is similar to the reduced growth response of the larvae (Clark et al., 2009; Sheppard Brennand et al., 2010) and juveniles (Byrne et al., 2014) of this species under anthropogenic acidification (i.e. direct bubbling of CO_2). In all treatments, reduction of A_T of the culture water limited the capacity of the system to buffer against increases in pCO_2 . Our results

highlight the concern that climate driven ocean acidification as an additional source of CO₂ will have significant impact on aquaculture of calcifying organisms, leading to reduced production and lower reproductive output by broodstock unless continual adjustments to water chemistry can be made cheaply by industry. Management of carbonate chemistry of culture water throughout the entire lifecycle will likely be required for efficient production of calcifying organisms and will become a more acute need as atmospheric CO₂ levels increase, a condition noted for culture of oysters in regions prone to upwelling of corrosive high CO₂ seawater (Barton et al., 2012).

The quantity and colour of sea urchin gonads are key determinants of market acceptance and value (Agatsuma et al., 2005; Pearce et al., 2002), and thus the economic viability of aquaculture ventures. We found that size, colour and nutritive condition of gonads depended on stocking density and seawater exchange rates. GIs were up to 89% higher in the low versus high density treatments, and 81% higher in the medium and high versus low exchange rate treatments. Moreover, both subjective and objective measures of gonad colour indicated that gonad quality was reduced by high stocking density and low seawater exchange rates. Studies that have examined the effects of stocking density on growth have found GI is reduced when sea urchins are kept at densities higher than used here (Juinio-Menez et al., 2008b; Siikavuopio et al., 2007a). Conversely, there was no effect of density on GI for other sea urchins cultured at similar or lower densities than used here (Christiansen and Siikavuopio, 2007; Gago and Luís, 2010; James, 2006), but these were all temperate species that may have inherently low growth and metabolic rates.

The growth rates of the *T. gratilla* in this study were similar to those reported for adults of this species in culture (Lawrence and Agatsuma, 2013; Seymour et al., 2013). Importantly, SGR and LGR were respectively 97% and 197% higher in the low density/high exchange rate treatment compared to the high density/low exchange rate treatment. Relative spine length

was up to 56% greater for sea urchins cultured at low versus high densities, and up to 70% greater at high versus low seawater exchanges. Our results contrast with other studies that have found no effect of density on the growth of adult sea urchins (Christiansen and Siikavuopio, 2007; James, 2006; Juinio-Menez et al., 2008b), possibly because growth rates were low or zero.

The lowest growth rates of *T. gratilla* occurred in high density and low seawater exchange rate treatments where pH, A_T and ΩCa were low, and pCO_2 was high. Confirming this observation, model selection using a modified Akaike's an Information Criterion (AICc) found growth (SGR, LGR) and gonad production (GI) were best correlated with ΩCa and pH (Table 2, Table S1), whilst consumption rates were correlated with ΩCa , pH and pCO_2 . It is well documented that 'acidified' seawater reduces post-settlement growth in sea urchins (Albright et al., 2012; Mos et al., 2015; Stumpp et al., 2012b; Wolfe et al., 2013); however, few studies have examined its effect on gonad size. In extremely low pH (6.98), gonad growth of the temperate *Strongylocentrotus droebachiensis* was 67% lower than controls cultured at ambient pH (8.10) (Siikavuopio et al., 2007b), and was reduced by 15% and 37% at pH 7.67 and pH 7.25 respectively compared to pH 7.98 (Stumpp et al., 2012b). In long-term experiments, there was little effect of moderate acidification (pH 7.6 – 7.8) on GI for sub-tropical/tropical sea urchins after six months, although gonad development was slowed (Kurihara et al., 2013; Uthicke et al., 2014). Other studies have found reduced egg production and gamete performance rather than examining gonad size (Dupont et al., 2013; Suckling et al., 2014; Uthicke et al., 2013).

Reduced ΩCa could have influenced the growth of both calcified (LGR, SGR) and non-calcified (GI) structures of *T. gratilla* by increasing the energy required to precipitate carbonate structures, reducing energy available for growth (Borja et al., 2009; Holtmann et al., 2013; Miles et al., 2007; Siikavuopio et al., 2007b). Reduced pH may slow growth, gonad

production and consumption rates by disrupting internal acid base balance and inducing hypercapnia (Hofmann and Todgham, 2010; Portner, 2012; Stumpp et al., 2012b). One mechanism used by sea urchins to counter decreases in internal pH is the active uptake of bicarbonates (Catarino et al., 2012; Collard et al., 2013; Moulin et al., 2015; Stumpp et al., 2012b) which is thought to be energetically costly (Catarino et al., 2012; Stumpp et al., 2012a) and reduces energy available for growth and reproduction (reviewed by Dubois, 2014). If sea urchins cannot compensate internal pH via uptake of bicarbonate and/or by dissolution of carbonate structures (Holtmann et al., 2013), hypercapnia may also reduce metabolic rates (Hofmann and Todgham, 2010; Portner, 2012; Stumpp et al., 2012b), reducing growth and consumption rates (Gillooly et al., 2001) and possibly gonad development. However in species acclimated or adapted to periodic hypercapnic or hypoxic conditions in their natural habitats, increased $p\text{CO}_2$ had generally little effect on metabolic rates (Stumpp et al., 2012b; Stumpp et al., 2011; Suckling et al., 2015). *T. gratilla* may be particularly susceptible to changes in pH because it generally does not experience hypercapnic or hypoxic conditions in natural habitats (Lawrence and Agatsuma, 2013) or because it is a fast-growing, tropical species with a high metabolic rate (e.g. Dy et al., 2002; Vink and Atkinson, 1985). Indeed, the short lifecycle of this species (often < 12 months, Ebert, 1982; Juinio-Menez et al., 2008a) may preclude long-term acclimation to low pH seen for perennial sea urchins (Kurihara et al., 2013; Uthicke et al., 2014).

Relative spine length of *T. gratilla* was highly correlated with pH and $p\text{CO}_2$. Low pH and high $p\text{CO}_2$ may have reduced spine length by reducing growth of spines (calcification) through the mechanisms described above. Another explanation could be that changes in pH/ $p\text{CO}_2$ decreased spine strength, making them more likely to break. Exposure to low pH and high $p\text{CO}_2$ reduces spine strength through dissolution in the sea urchins *S. droebachiensis* (Holtmann et al., 2013; Siikavuopio et al., 2007b) and *Lytechinus variegatus* (Albright et al.,

2012). Spines may be particularly susceptible to dissolution under low pH or high $p\text{CO}_2$ conditions because they are more exposed to seawater (Holtmann et al., 2013), whilst the plates of the test are thought to be partially protected by increases in extracellular bicarbonate in the perivisceral coelomic fluid (Holtmann et al., 2013; Stumpp et al., 2012b). Spine strength may have played a role in determining spine lengths in this study because relative spine length was reduced at high densities where ‘collisions’ between urchins may be high (Siikavuopio et al., 2007a), but this cannot be confirmed as the number of spine fragments in treatments was not measured.

Differences in the seawater carbonate chemistry among treatments were most likely generated by release of respiratory CO_2 by *T. gratilla* and the uptake of bicarbonates (Mos et al., 2015). Modelling using literature derived constants showed that respiratory CO_2 production by *T. gratilla* (Table S4) would have been sufficient to generate a large proportion of $p\text{CO}_2$ measured in treatments. Decomposition of faeces and respiration by the *Sargassum* sp. diet are unlikely to have substantially contributed to increased $p\text{CO}_2$ (Mos et al., 2015). Reductions in A_T were most likely caused by the uptake of bicarbonates for calcification and internal pH buffering by the sea urchins, similar to juvenile *T. gratilla* (Mos et al., 2015).

Our results indicate that growth and reproduction of *T. gratilla* may be vulnerable to acidification in natural systems associated with increases in atmospheric CO_2 (Caldeira and Wickett, 2005; IPCC, 2013), although it is unknown whether the natural occurrence of *T. gratilla* in high densities (e.g. Valentine and Edgar, 2010) or areas with low seawater exchange (e.g. Lawrence and Agatsuma, 2013) would create similar localised changes in A_T and other carbonate parameters as measured in our experiment. The generally stable pH and $p\text{CO}_2$ concentrations recorded in treatments in this study correspond to predicted future levels ($\sim 0.3 - 0.5$ pH unit decrease below ambient (Caldeira and Wickett, 2005; IPCC, 2013)).

Growth (SGR, LGR, relative spine length) and gonad production (GI) were reduced in *T.*

gratilla cultured under acidified conditions. As GI is directly related to the reproductive output of sea urchins (e.g. Luis et al., 2005), our results suggest that acidification reduces reproductive capacity of calcifiers (Dupont et al., 2013; Uthicke et al., 2013) by directly reducing gonad growth and gamete production. Gamete development in sea urchins depends on the nutrients accrued by the nutritive cells (Byrne, 1990) and the poor condition of these cells in gonads from the high density/low exchange rate treatment would also compromise reproduction (also see Kurihara et al., 2013; Uthicke et al., 2014). Additionally, we found gonad colour was poorest in treatments where pH was lowest and $p\text{CO}_2$ highest. Gonad colour is determined by carotenoids, and is therefore indicative of the amount of carotenoids available for maternal provisioning of gametes (Griffiths and Perrott, 1976). Maternally-derived carotenoids are linked to successful larval development, survival and metamorphosis (de Jong-Westman et al., 1995). Thus decreased carotenoid provisioning could explain lower survival rates of sea urchin larvae from adults acclimated to high $p\text{CO}_2$ conditions (Dupont et al., 2013; Suckling et al., 2015; Uthicke et al., 2013).

This study suggests that aquaculture of adult marine calcifiers may be highly vulnerable to the effects of increased atmospheric CO_2 concentrations. *T. gratilla* significantly reduced the buffering capacity (A_T) of their culture water likely through uptake of carbonates as they grew, thereby decreasing the pH and ΩCa for a given $p\text{CO}_2$. In this situation, even small increases in CO_2 can have a comparatively large impact on carbonate system (also see Mos et al., 2015). Furthermore, low A_T may prevent calcifying organisms from acclimatising to low pH conditions by inhibiting the availability of bicarbonate for internal pH balance. As the ocean is on a trajectory of increasing $p\text{CO}_2$, direct management of culture water carbonate chemistry will be required to maximise production in intensive cultures of larval (Barton et al., 2012), juvenile (Mos et al., 2015) and adult marine calcifiers.

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Supporting Information

for

Biogenic acidification reduces sea urchin gonad growth and increases susceptibility of aquaculture to ocean acidification

*Benjamin Mos^{*1}, Maria Byrne² and Symon A. Dworjanyn¹*

¹ National Marine Science Centre, Southern Cross University, Coffs Harbour, New South Wales, Australia.

² Schools of Medical and Biological Sciences, University of Sydney, Sydney, New South Wales, Australia

* Corresponding author: Benjamin Mos

Phone: (02) 6648 3900

Address: PO Box 4321, Charlesworth Bay, Coffs Harbour NSW 2450, Australia

E-mail address: benjamin.mos@scu.edu.au

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FIGURE

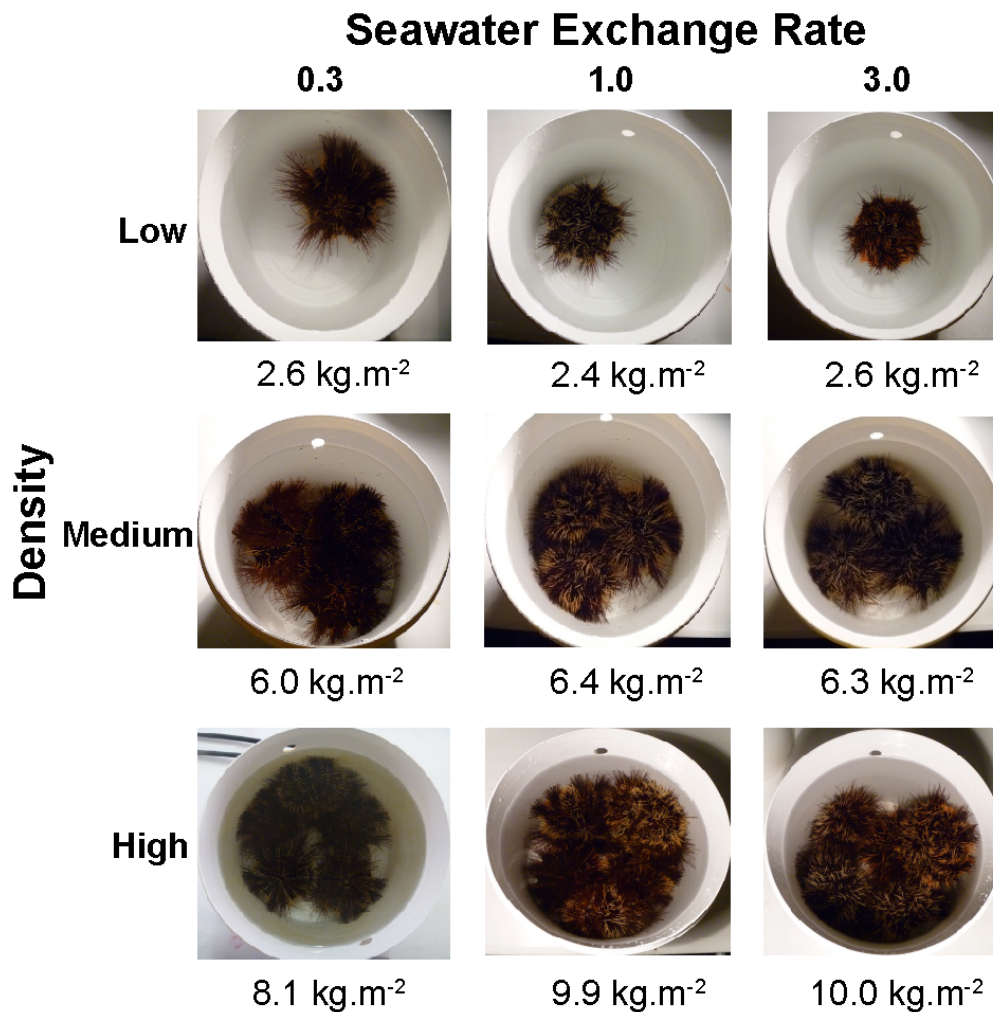


Figure S1: Representative size and densities of adult *Tripneustes gratilla* after six weeks in a flow-through culture system under nine density and seawater exchange rate regimes. Low, Medium and High densities represent initial densities of 9, 26 and 43 individuals.m⁻² respectively, equivalent to 1, 3 and 5 individuals per container respectively. Seawater exchange rates refer to 0.3, 1.0 and 3.0 exchanges.hr⁻¹ of replicate container (4.0 L). Numbers below figures represent final mean densities in kg.m⁻² of total surface area.

TABLES

Table S1: Distance-based linear model (DISTLM) marginal tests for Specific Growth Rate (SGR), Linear Growth Rate (LGR), relative spine length (the ratio of spine length to test diameter: RSL), gonad index (GI) and consumption rates of *Tripneustes gratilla* and selected environmental variables. Selection procedure: BEST, selection criterion: AICc. ‘Prop.’ is the proportion of variation in the dataset explained by the variable. Variables with $P > 0.01$ are bold, and were excluded from further analysis. ΩCa = Calcite Saturation State, DO = dissolved oxygen, $p\text{CO}_2$ = dissolved carbon dioxide, Temp. = Temperature.

Dataset	Variable	SS (trace)	F	P	Prop.
SGR	ΩCa	81.58	20.18	0.0001	0.32
	$p\text{CO}_2$	151.62	62.81	0.0001	0.59
	pH	125.12	41.29	0.0001	0.49
	DO	151.74	62.94	0.0001	0.59
	Salinity	15.86	2.85	0.1014	0.06
	Temp.	10.991	1.93	0.1731	0.04
LGR	ΩCa	12.17	17.57	0.0003	0.29
	$p\text{CO}_2$	17.41	30.52	0.0001	0.42
	pH	16.59	28.13	0.0001	0.4
	DO	14.87	23.61	0.0001	0.35
	Salinity	3.18	3.53	0.0623	0.08
	Temp.	4.47	5.13	0.0303	0.11
RSL	ΩCa	311.77	83.72	0.0001	0.66
	$p\text{CO}_2$	216.05	36.31	0.0001	0.46
	pH	318.22	89.03	0.0001	0.67
	DO	115.56	13.95	0.0001	0.24
	Salinity	253.06	49.72	0.0001	0.54
	Temp.	202.33	32.28	0.0001	0.42
GI	ΩCa	54.56	43.96	0.0001	0.51
	$p\text{CO}_2$	46.75	32.87	0.0001	0.43
	pH	55.92	46.23	0.0001	0.52
	DO	32.8	18.78	0.0003	0.3
	Salinity	28.37	15.34	0.0005	0.26
	Temp.	26.92	14.29	0.0008	0.25
Consumption	ΩCa	2465.3	55.5	0.0001	0.56
	$p\text{CO}_2$	2369.7	50.81	0.0001	0.54
	pH	2892.2	83.85	0.0001	0.66
	DO	1792.1	29.83	0.0001	0.41
	Temp.	1000.3	12.74	0.001	0.23

Table S2: PERMANOVA analyses examining the effects of seawater exchange rate and density of *Tripneustes gratilla* on pH, partial pressure of dissolved carbon dioxide ($p\text{CO}_2$), calcite saturation state (ΩCa) and total alkalinity (A_T) over six weeks in a flow-through seawater system. Er = Exchange Rate, De = Density. Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

	Source	df	MS	F	P	Post-hoc tests	
						Interaction	Interaction
pH	Exchange Rate	2	0.24	404.29	0.0001	low (3.0 > 1.0 > 0.3)	0.3 (low > medium > high)
	Density	2	0.22	364.55	0.0001	medium (3.0 > 1.0 > 0.3)	1.0 (low > medium > high)
	Er × De	4	0.01	20.73	0.0001	high (3.0 > 1.0 > 0.3)	3.0 (low > medium > high)
	Residual	36	5.96e-4				
						Interaction	Interaction
$p\text{CO}_2$	Exchange Rate	2	66455	53.48	0.0001	low (3.0 = 1.0 < 0.3)	0.3 (low < medium < high)
	Density	2	2.46e5	197.97	0.0001	medium (3.0 > 1.0 = 0.3)	1.0 (low < medium < high)
	Er × De	4	11967	9.63	0.0002	high (3.0 < 1.0 < 0.3)	3.0 (low < medium < high)
	Residual	36	1242.7				
						Interaction	Interaction
ΩCa	Exchange Rate	2	33.90	1783.5	0.0001	low (3.0 > 1.0 > 0.3)	0.3 (low > medium > high)
	Density	2	14.86	781.99	0.0001	medium (3.0 > 1.0 > 0.3)	1.0 (low > medium > high)
	Er × De	4	0.54	28.46	0.0001	high (3.0 > 1.0 > 0.3)	3.0 (low > medium > high)
	Residual	36	0.01				
						Interaction	Interaction
A_T	Exchange Rate	2	2.81e6	541.22	0.0001	low (3.0 > 1.0 > 0.3)	0.3 (low > medium = high)
	Density	2	2.41e5	46.39	0.0001	medium (3.0 > 1.0 > 0.3)	1.0 (low > medium > high)
	Er × De	4	57444	11.05	0.0001	high (3.0 > 1.0 > 0.3)	3.0 (low > medium = high)
	Residual	45	5196.7				

Table S3: Mean seawater physico-chemical parameters measured in nine seawater exchange rate by density treatments. Temperature, dissolved oxygen (DO) and salinity are averaged across replicate within each treatment (n = 5), whilst weekly measures for nitrate concentration were considered as replicates in each of the nine treatments (n = 6). Ambient values were obtained from the seawater supplied to the experiment. Values in parentheses are SE.

Treatment		Seawater Physico-chemical Parameters			
Exchange Rate (exchanges.hr ⁻¹)	Density (urchins.replicate ⁻¹)	Temperature (°C)	DO (mg.L ⁻¹)	Salinity (ppt)	Nitrate (mg.L ⁻¹)
0.3	5	26.5 (0.1)	6.20 (0.02)	36.4 (0.0)	0.01 (0.00)
0.3	10	26.5 (0.1)	6.07 (0.03)	36.5 (0.0)	0.00 (0.00)
0.3	15	26.5 (0.1)	5.95 (0.04)	36.5 (0.0)	0.00 (0.00)
1.0	5	26.8 (0.1)	6.25 (0.01)	35.9 (0.1)	0.13 (0.04)
1.0	10	26.8 (0.1)	6.09 (0.01)	35.9 (0.0)	0.07 (0.03)
1.0	15	26.7 (0.1)	6.01 (0.01)	35.9 (0.0)	0.03 (0.01)
3.0	5	27.4 (0.1)	6.20 (0.02)	35.9 (0.0)	0.32 (0.05)
3.0	10	27.4 (0.1)	6.09 (0.01)	35.9 (0.0)	0.28 (0.06)
3.0	15	27.4 (0.1)	6.02 (0.03)	35.9 (0.0)	0.26 (0.05)
Ambient		27.4 (0.1)	6.41 (0.01)	36.1 (0.1)	0.33 (0.06)

Table S4: Estimated CO₂ production by *Tripneustes gratilla* during week six of the density and seawater exchange rate experiment. Values calculated utilising regression coefficient (b) and constant (K) values for *T. gratilla*¹ under similar temperature and salinity conditions to this study for the equation $M = KW^b$ relating body mass and metabolic rate². Dry weights for the sea urchins were calculated from wet weights using a wet:dry weight ratio of 0.1359 (Mos, unpublished data, n = 131). Note: Constant values are for *T. gratilla* within ambient seawater. Oxygen uptake is reduced for echinoderms exposed to low pH³, and therefore CO₂ production values for the 0.3 exchanges.hr⁻¹ treatments are likely overestimates. To assist with estimating CO₂ production by the urchins, CO₂ values in the treatments were converted from µatm to µmol using the Ideal Gas Law.

Treatment		CO ₂ in Treatments		Estimated CO ₂ Produced by Sea Urchins		
Seawater Exchange Rate (exchanges.hr ⁻¹)	Density (individuals. replicate ⁻¹)	Mean CO ₂ (µmol) (Calculated from pH & A _T)	CO ₂ Required to Maintain CO ₂ Above Ambient (43.36) (µmol.hr ⁻¹ .replicate ⁻¹)	Mean Urchin Mass (g. individual ⁻¹) Dry Weight	Respiration Rate (µmol CO ₂ . g ⁻¹ . hr ⁻¹) (Calculated)	CO ₂ Produced (µmol.hr ⁻¹ . replicate ⁻¹) (Calculated)
0.3	1	62.05	6.23	33.25	0.45	14.96
0.3	3	110.98	20.29	25.44	0.63	48.08
0.3	5	125.96	24.78	20.53	0.83	85.20
1.0	1	53.01	9.65	29.86	0.51	15.23
1.0	3	81.19	37.83	26.70	0.59	47.26
1.0	5	107.41	64.05	24.91	0.65	80.96
3.0	1	50.89	22.59	32.70	0.46	15.04
3.0	3	71.83	85.41	26.55	0.60	47.79
3.0	5	83.26	119.7	25.22	0.64	80.70

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